UG (NEP) 3rd SEMESTER
Applied Computing – Digital Electronics
(ACP322N)

No. of Questions: 09 Session: December 2025
Time Allowed: 2:00 Hours Max. Marks: 100 Min. Marks: 36

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

i. Define logic level, pulse, and clock in digital waveforms?

ii. Convert $(1100101)_2$ to decimal?

iii. Represent −25 in 8-bit 2’s complement?

iv. What are don’t-care conditions in K-maps?

v. Define encoder and decoder?

vi. State one difference between microprocessor and microcontroller?

vii. Using DeMorgan’s laws, simplify the Boolean expression shown in the question paper: complemented AB plus the complemented product of complemented A and complemented B.

viii. What is DMA in I/O and why is it used?

Section B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

2. Convert $(3A7)_{16}$ to binary and octal; show steps?

OR

Convert $(753.12)_{10}$ to binary and octal; show steps?

3. From the truth table of a 2-to-1 MUX, derive its Boolean expression and draw a gate-level circuit.

OR

Minimize F(A, B, C, D) = Σm(0, 2, 5, 7, 8, 10, 13, 15) using a 4-variable K-map?

4. Implement XOR using only NAND gates; show logic steps?

OR

Differentiate latch vs flip-flop; explain edge-triggered D flip-flop with a timing sketch?

5. Compare synchronous vs asynchronous counters; design a 3-bit synchronous up counter with JK flip-flops?

OR

Design a full adder?

Section C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain Boolean algebra laws and DeMorgan’s theorems with examples?

7. Discuss SOP and POS forms; map a 3-variable function to K-map and compare minimized SOP vs POS?

8. Describe shift registers with timing; design a 4-bit serial-in serial-out register?

9. Explain CPU organization: ALU, control unit, register set, buses; outline instruction execution cycle with timing?